time it swallows something, it makes sense that the entropy is connected to the size of the black hole itself—specifically, it's related to the total surface area of the event horizon. Throw a refrigerator into a black hole, and the mass increases by the mass of the refrigerator, which increases the horizon size and therefore the surface area.[fn10]

The fact that you can't have entropy without having a temperature means that black holes have to be radiating something (particles and radiation, specifically). And the only place they *can* radiate from is at or just outside the event horizon, since we still can't have anything coming out once it's gone in. So something weird has to be happening around there.

Fortunately, if we need weirdness in physics, we can always rely on the quantum realm to serve us up something good. In this case, Hawking made use of the quantum weirdness of *virtual particles*—pairs of positive- and negative-energy particles popping into and out of existence from the vacuum of space itself.[fn11] The idea is that this spacetime popcorn is happening *all the time*, everywhere, but usually it has no effect on anything because the two particles will appear and immediately annihilate against one another, both going back to being nothing again. But, Hawking said, near a black hole, you could have a situation where the negative-energy virtual particle falls past the horizon, leaving the positive-energy virtual particle so bereft that it becomes real and wanders away. The mass of the black hole would reduce a little as it absorbs that bit of negative energy, and the same amount of positive energy would appear to radiate off the black hole's horizon. Because these virtual particles are always popping up everywhere in space, any black hole that's not actively pulling in matter from its environment should be gradually bleeding off mass through this evaporation process all the time.

As complicated as this might sound, it's still a vastly simplified picture, meant to capture just the basic idea without getting *too* technical, and it's an explanation that's used all the time. But it has never been particularly satisfying to me, since it seems to require the negative-energy particles to preferentially fall toward the hole, and the positive-energy ones to be traveling away from the black hole with enough energy to escape. It turns out that despite talking in these terms for popular audiences, Hawking never really wanted this explanation to be taken literally, and the real explanation involves calculating wave functions and the scattering that the waves experience in the vicinity of a black hole. I can't really get into it without a